# Revisiting Division of Labor

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# Thoma’s Critique

Last time we saw that in Weisberg and Muldoon's model, mavericks outperform followers, and mixed populations of mavericks and followers outperform pure follower populations.

Weisberg and Muldoon concluded that mixed populations, with both followers and mavericks, are ideal for many research domains.

But Johanna Thoma (2015) argues that their model doesn't actually show this. She identifies two problems with the model and builds an alternative that gives a more convincing case for diversity of research strategy.

# What Did Weisberg and Muldoon Actually Show?

Thoma points out that in the total progress measure (Figure 11 of their paper), homogeneous populations of mavericks **outperform** all mixed populations. More mavericks always means more total progress.

So if you hold population size fixed at 400 and vary the ratio of mavericks to followers, the all-maverick community does best. The mixed population result only appears when you add mavericks to a fixed number of followers.

Weisberg and Muldoon's argument for mixed populations rests on the claim that being a maverick is costlier. But they don't model those costs. The case for diversity ends up relying on an assumption external to the model.

# Two Kinds of Division of Labor

Thoma makes a useful distinction between two ways of dividing up labor in a scientific project.

- Division of *research approach* (what topic you work on, method you use, etc.)
- Division of *research strategy* (how you choose your topic, method, etc.)

Kitcher and Strevens were interested in the first kind: how do we get scientists to spread across different projects?

Weisberg and Muldoon are interested in the second: does a community benefit from having some scientists who explore and others who follow?

These are different questions, and it would be a bit surprising if they were modeled the same way..

## Problem 1: The Follower Rule

Thoma's first criticism is that the follower strategy is kind of dumb.

In Weisberg and Muldoon's model, followers move to the most significant previously visited patch in their neighborhood.

This means they often, almost always, end up on patches that have already been explored by someone else.

In the model, visiting a patch someone else has already visited contributes nothing to epistemic progress. So followers spend much of their time doing what amounts to pointless duplication.

This is unrealistic.

# Problem 2: Local Movement

The second criticism is that agents can only move one patch per round. Thoma argues this implies an extreme degree of inflexibility and ignorance.

Given the fine-grained interpretation of the landscape (one patch per round, 101 by 101 grid), local movement means scientists can't become aware of research happening more than a few months away from their own.

If each round is about 2 months, a scientist with range 1 can only see what's happening within 2 months of their current work.

There is an important general point here. The model is a giant metaphor. But sometimes what something in the model is a metaphor for changes when we look at different applications or aspects. This can cause problems.

# Thoma’s Alternative Model

Thoma builds an alternative model keeping the same landscape but changing the agent strategies. She introduces **Explorers** and **Extractors**.

Both Explorers and Extractors avoid mere duplication: they never visit a patch that someone else has already visited. This is probably the single biggest change between the original model and hers.

The difference between Explorers and Extractors is in how they respond to prior discoveries in their neighborhood.

# The Explorer Rule

When an Explorer sees previously visited patches nearby, it moves to the unvisited patch that is **farthest** from those visited patches. It heads away from where others have been.

When in uncharted territory (no previously visited patches nearby), Explorers behave like hill climbers: they move forward and keep going if significance is increasing, back up and try a new direction if it's decreasing.

Explorers seek out new ground. They want to be doing something that nobody else is doing.

Remember that the climbers, the controls in Weisberg and Muldoon's model, did fine. They were painfully slow, at least without supplementation, but they got there.

Feels like a sensible thing to check whether climbing in sparsely explored territory is a good way to contribute to the collective knowledge gain.

# The Extractor Rule

When an Extractor sees previously visited patches nearby, it moves to the unvisited patch that is **closest** to the best previous discovery. It gravitates toward areas that look promising based on prior work, but doesn't duplicate what's already been done.

In uncharted territory, Extractors also behave like hill climbers.

Extractors are like what most normal scientists are like. They learn their field, find out what questions look like they can be attacked with the skills they learned, and attack them. This is good; this is what we want most people to be doing.

# Variable Range of Movement

Thoma also models what happens when we change the range that agents can move. Instead of being restricted to adjacent patches (range 1), agents can perceive and move to patches within radius $r$, where this is a free parameter in the model.

This controls how flexible and informed scientists are. At range 1, scientists are extremely short-sighted. At range 10, they can see a substantial part of the landscape. At global range, they know about all previous discoveries.

The range parameter turns out to matter a great deal for whether diversity of strategy is beneficial.

It would be useful to think about what it would mean to have diversity of ranges *within* a single model.

# Results

With local movement, Thoma's results resemble Weisberg and Muldoon's. Homogeneous groups of Explorers outperform both homogeneous groups of Extractors and mixed groups, at least until very late in the simulation.

After 50 rounds, all-Explorer groups have discovered 15% of total significance, while 50/50 mixed groups have only discovered 8%. The mixed group does eventually overtake the all-Explorer group, but only after 90% of the significance has been found.

So with local movement, there is no advantage to diversity of research strategy. O'Connor's earlier point about the weakness of follower-type agents under local movement holds for Thoma's model too.

# Medium Range Movement (Range 3-10)

When the range of movement increases, the results flip. From range 3 onward, mixed populations outperform homogeneous populations of either type.

At range 10, the advantage of mixed groups is substantial and statistically significant at every time point.

A secondary benefit of mixing: the standard deviation of outcomes drops sharply. Homogeneous Extractor groups have a standard deviation of 12.5 percentage points of coverage after 150 rounds; mixed groups with 40% Explorers have a standard deviation of only 3.1 percentage points.

Mixed communities are not just better on average, they are more reliable.

With local movement, Extractors can barely see what Explorers have done. An Explorer might pass through a patch right next to an Extractor, but if the Extractor's perception range is only 1, it probably won't notice.

With medium-range movement, Extractors can actually find the trails Explorers have blazed and follow them to productive areas. The two strategies become genuinely complementary: Explorers find the hills, and Extractors work them systematically.

Thoma's interpretation: diversity of research strategy is beneficial when scientists are not too inflexible and not too ignorant of work outside their immediate area. The more flexible and informed scientists are, the more beneficial diversity becomes.

At global range, diversity is still beneficial, but overall progress is **slower** than in the medium range case. Here we get back to a Zollman-type result.

When Extractors can see the entire landscape, the less significant peak gets neglected as soon as the more significant one is found.

In these models, certain degree of ignorance can be epistemically productive. If scientists had perfect information, they would all flock to the highest peak. Limited information keeps some of them working on the second-best peak, which is still worth exploring.

# Who Would Choose to Be an Explorer?

There is a residual problem. In the mixed-population simulations, Extractors are about four times as productive as Explorers at medium range.

If scientists care about their own productivity (as credit economy models suggest), few would choose the Explorer strategy.

In the model, it's good to **have** Explorers, but it's not so good to **be** an Explorer. Most exploration fails; but the ones that succeed encourage a flood of followers.

# Who Would Choose to Be an Explorer?

Thoma's response draws on the credit economy literature. Social and financial rewards tend to go disproportionately to novel work. Grant agencies prize novelty. Fame accrues to those who open new areas. The ERC in the EU has made high-risk "frontier research" its core mission.

So non-epistemic rewards might sustain a population of Explorers even though the strategy is less epistemically productive for the individual. This connects back to the Kitcher/Strevens point that credit incentives can maintain a socially beneficial division of labor.

# Summing Up

# What the Landscape Models Show

Thoma's model gives a more convincing case for diversity of research strategy than Weisberg and Muldoon's.

Division of labor between explorers and extractors is beneficial, but only when scientists have enough flexibility and awareness to actually respond to each other's work.

The overall lesson from this literature: cognitive diversity in scientific communities can promote progress, but the specific mechanisms matter. Agent-based models help us see when and why diversity helps, and also when modeling choices might be driving the results rather than genuine features of scientific communities.

# Stepping Back

# Themes From the Last Few Weeks

We've now spent several weeks on formal models of scientific communities:

- Zollman on networks,
- Weatherall/O'Connor on polarization models,
- The Matthew effect and citation gap models,
- And now the Weisberg-Muldoon and Thoma landscape models.

These are very different models, but a few common themes are worth pulling out, both for thinking about the essay questions and for thinking about agent-based modeling more generally.

In almost every case we've looked at, we start from a situation where we already know there are forces pulling in opposite directions.

It is good for scientists to share evidence; but it is also good for some scientists to keep working on a theory even when the current evidence looks bad. Which force dominates?

It is good to have people piling onto the most promising approach; but bad for everyone to work on the same thing. Which force dominates?

It is good to reward success; but that same reward structure can lead to runaway reputation effects and unequal credit. Which force dominates?

# Toy Models Can Clarify the Trade-Off

Often, the interesting work the model does is telling us *when* one force wins and when the other does.

The Zollman effect says: when the signal is weak and populations are small, the information-preservation force beats the information-sharing force.

Rosenstock et al. then tell us that when the signal is moderately strong, it's the other way around.

Thoma's paper is a nice example of something we've seen repeatedly: modeling arguments often trade on slippage between two things that look the same but aren't.

In Thoma's case, "diversity of research" is ambiguous between diversity of *approach* (what you work on) and diversity of *strategy* (how you decide what to work on).

Kitcher and Strevens are modeling the first; Weisberg and Muldoon claim to show something about the second. The conclusions don't transfer automatically.

We saw a similar ambiguity in the network models.

"Connectedness" in Zollman's model means whether you see another scientist's experimental results.

"Connectedness" in the landscape models means whether you can perceive what another scientist is working on.

A small-world network is not necessarily a small-world geography.

# Why This Matters

When we translate model results into claims about real science, the translation step is doing a lot of work.

Is a densely connected Zollman network more like a field with open preprint servers, or more like a field where everyone goes to the same conference?

These give very different real-world predictions, and the model alone doesn't tell us which translation is right.

Part of the discipline of reading this literature is keeping track of what each piece of the model is supposed to represent, and whether the argument still goes through when we swap in a different interpretation.

## Theme 3: When Is Unrealism a Fair Complaint?

Every model we've looked at is unrealistic in many ways. The hard question is which kinds of unrealism matter.

Thoma's complaint that Weisberg and Muldoon's followers do something no real scientist would do, wander onto already-visited patches to no purpose, seems fair.

The unrealism here is driving the result that followers are unproductive.

By contrast, complaining that Weisberg and Muldoon's scientists are ants on a lumpy donut seems like a bad complaint.

# A Rough Heuristic

A useful question to ask about any modeling assumption: *if I changed this, would the headline result change?*

If yes, then the assumption is doing argumentative work and needs defending. That's what Rosenstock et al did on Zollman's model, and what Thoma does on Weisberg and Muldoon's model.

If no, then the assumption is a simplification and the unrealism is probably fine.

In practice, we use **robustness** checks as a test for whether the result is sufficiently realistic.

# Looking Ahead

Next week we'll talk about the essay questions.

They all pick up some of these themes, about what the model shows, what the assumptions do, and how robust the results are.

Think about which of these questions most appeals to you, and come with questions.